

BanLegit-Cite: Sequential Single-Owner Execution Plan

Minimum-Coordination Architecture for Two Researchers

Design principle: Each phase has exactly one owner who makes all in-phase decisions and produces one exit deliverable package. The other researcher does not co-work the same tasks inside that phase — they either do genuinely independent prep work that has zero dependency on the active phase, or they wait. Coordination only happens at fixed handoff gates, never mid-phase.

WHY THIS PHASE ASSIGNMENT IS OPTIMAL

Ownership alternates between two skill profiles, not between two people arbitrarily:

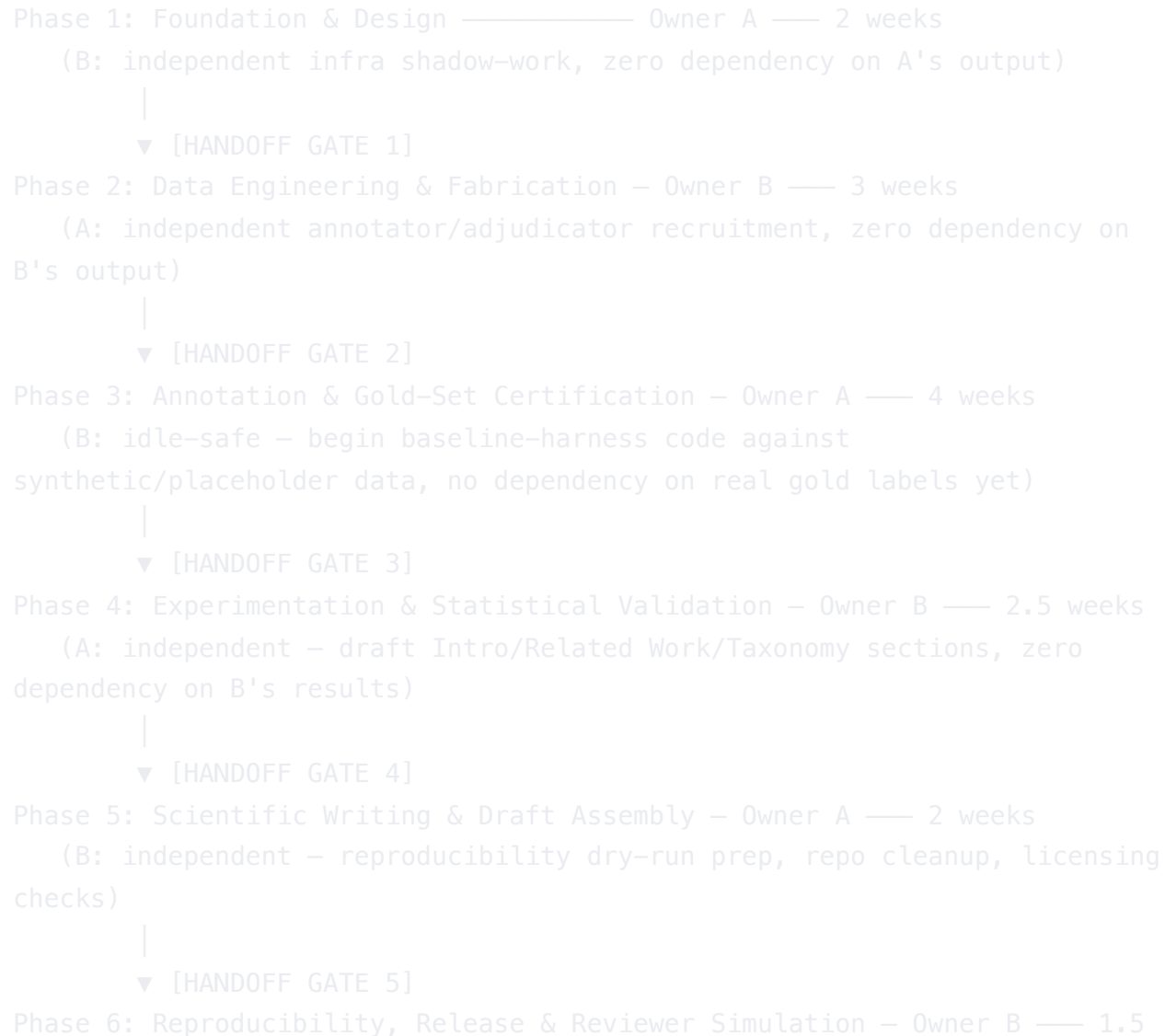
- **Researcher A = Judgment & Narrative Owner** — owns phases requiring legal-domain judgment, taxonomy/annotation decisions, and coherent scientific narrative (Foundation, Annotation Certification, Writing).
- **Researcher B = Systems & Execution Owner** — owns phases requiring engineering throughput, pipeline construction, and statistical execution (Data Engineering, Experimentation, Reproducibility/Release).

This is not a 50/50 arbitrary split — it's a skill-matched split, which matters because single-owner phases only work if the owner can make *every* decision inside the phase without needing to consult the other person. If ownership were assigned arbitrarily, the owner would constantly hit judgment calls outside their strength and either stall or make weak calls. Matching phase-type to skill-strength is what makes true single-ownership viable at all.

The alternation also solves the idle-time problem inherent to strict sequencing: while A is deep in a judgment-heavy phase, B is never blocked — B is either doing independent shadow prep (Phase 1) or has already handed off and moved to their own next-phase

prep. The schedule below is explicitly designed so neither researcher is ever purely waiting with nothing productive to do, without violating the "no shared mid-phase work" rule.

PHASE MAP (OVERVIEW)



weeks



▼ [HANDOFF GATE 6]

Phase 7: Final Joint Review & Submission — JOINT (unavoidable) — 0.5 weeks

Total: ~15.5 weeks. Every arrow is a hard gate — the next phase does not start until the exit criteria below are met and a handoff package is delivered.

PHASE 1 — FOUNDATION & RESEARCH DESIGN

- **Objective:** Establish and freeze everything the rest of the project is built on: novelty validation, gap statement, RQ/H1-H3, dataset scope, and taxonomy v1.
- **Single owner: Researcher A**
- **Deliverables:**
 - Novelty validation report + competitor comparison table
 - Frozen gap statement
 - Frozen RQ + H1-H3 with metric mapping
 - Dataset scope document (Acts, courts, date range) with confirmed institutional access
 - Taxonomy v1 (pilot-tested, $\kappa \geq 0.6$, with decision tree)
 - Annotation guideline v1 draft
- **Required AI agents:** A2 (Literature Review), A3 (Novelty Verification), A4 (Research Gap), B1 (Benchmark Design — task/metric skeleton only at this stage), B2 (Citation Taxonomy), B3 (Dataset Discovery), C3 (Annotation Guideline, draft only).
- **AI tools:** Connected Papers + ResearchRabbit (seed discovery), Semantic Scholar/OpenAlex API (bulk metadata via A2), Consensus (adversarial claim-checking via A3), Litmaps (set up monitoring now, runs through the whole project),

NotebookLM (grounded synthesis across the 6 core competitor papers), Zotero (bibliography, established as canonical from day one).

- **Human supervision checkpoints:**
 - A personally reads all 6 core competitor papers in full before accepting A3's novelty report.
 - A personally signs the gap statement and RQ/H freeze in the decision log.
 - Pilot annotation round (A recruits 1-2 pilot annotators, e.g., a law student, for a 20-instance test) run before taxonomy is frozen.
- **Entry criteria:** Project kickoff; repo initialized; Zotero library created.
- **Exit criteria:** Novelty table complete and A-signed; RQ/H frozen; taxonomy pilot $\kappa \geq 0.6$; dataset scope document complete with confirmed access to all sources (this is the hardest external dependency — do not let this slip past Phase 1, since B's entire Phase 2 depends on it).
- **Estimated duration:** 2 weeks.
- **Git branch/workflow:** `phase1-foundation` branch, owned solely by A; B does not commit here. Merge to `main` only at the handoff gate, after B's review (B reviews for feasibility/clarity, not content authority — a fast sanity-check PR review, not a content negotiation).
- **Handoff artifacts to Phase 2:**
 - `data-scope.md`, `taxonomy_v1.md` + decision tree, `guideline_v1.md`, `novelty_report.md`, `gap_statement.md`, `rq_hypotheses.md` — all merged into `main`.
 - A 45-60 minute recorded or live handoff walkthrough where A explains taxonomy edge cases and scope rationale directly to B, since B will be building the scraper/fabrication pipeline against this without further access to A during Phase 2.

What B does during Phase 1 (independent shadow-work, not a shared task): repo scaffolding, CI setup, DVC/W&B configuration, Label Studio installation and generic

config, scraper framework skeleton (not yet pointed at real sources, since scope isn't final), Claude Code project-instructions file (`CLAUDE.md`) drafted from the Operating Manual. None of this depends on A's output content, only on the fact that a project exists — this is the safe overlap.

PHASE 2 — DATA ENGINEERING & FABRICATION PIPELINE

- **Objective:** Build and execute the full data-construction pipeline: scrape/extract the real corpus, generate the fabricated corpus, load everything into the annotation platform ready for human annotation.
- **Single owner: Researcher B**
- **Deliverables:**
 - Raw corpus (1,200–1,500 real citation instances) with full provenance metadata, $\geq 95\%$ spot-check pass rate
 - Fabricated corpus (batch-generated, batch-approved) matching the real corpus 1:1 by taxonomy category
 - Both loaded into Label Studio, ready for annotator assignment
 - Source-repository catalog with confirmed copyright status per source
- **Required AI agents:** B3 (Dataset Discovery — finalizes source catalog), C1 (Data Collection), C2 (Fabrication Generation).
- **AI tools:** Claude Code (scraper/OCR pipeline construction, fabrication batch scripts), DVC (corpus versioning from first commit), Gemini CLI (only if scanned-PDF OCR volume is high enough to warrant it).
- **Human supervision checkpoints:**
 - B personally spot-checks 5% of raw extractions against original sources before accepting the corpus.
 - **Fabrication batch review cannot be done by B alone** — this is the one mandatory touchpoint with A inside this phase (see note below), since

content-quality judgment on fabrications is a legal/linguistic call, not an engineering one.

- **Entry criteria:** Phase 1 handoff package received and walkthrough completed; scraper framework already built (from B's Phase 1 shadow-work).
- **Exit criteria:** Raw + fabricated corpora complete, provenance-verified, batch-approved (<10% rejection rate per category), loaded into Label Studio and assigned to annotators.
- **Estimated duration:** 3 weeks.
- **Git branch/workflow:** `phase2-data-engineering` branch, owned by B. A does not commit here. DVC-tracked data commits are frequent and small (per-batch), so any failure is isolated to a small diff, not a large irreversible corpus.
- **Handoff artifacts to Phase 3:**
 - Fully loaded Label Studio project, ready to assign
 - `data_catalog.md` (source, access method, copyright status per source)
 - Batch-approval logs (so A can see the fabrication-generation history without re-deriving it)
 - A 30-45 minute handoff walkthrough on any data-quality caveats (e.g., OCR-flagged instances, categories with tighter/looser fabrication difficulty than planned)

Exception to strict single-ownership — the fabrication batch-review touchpoint: the Operating Manual requires human content-review of each fabrication batch before the next unlocks (C2's constraint). To avoid turning this into constant mid-phase back-and-forth, **batch it:** B runs the full generation pipeline for one category at a time and packages 3-4 batches together; A reviews that package in a single 1-2 hour session per category (not per-batch), keeping total A-touchpoints during Phase 2 to roughly 5-6 short sessions (one per category) rather than dozens of small interruptions. This preserves the "minimize communication overhead" goal while still honoring the non-negotiable human-verification rule from the Operating Manual.

What A does during Phase 2 (independent, zero dependency on B's pipeline output): recruit and confirm 2–3 annotators (law students) and the senior adjudicator (practicing lawyer or faculty); finalize annotator training materials and the Label Studio annotation instructions UI text; run a second, larger pilot round with the recruited annotators (using Phase 1's placeholder examples, not requiring Phase 2's real corpus) to further stress-test the guideline before full annotation begins in Phase 3.

PHASE 3 — ANNOTATION & GOLD-SET CERTIFICATION

- **Objective:** Execute full double-annotation, compute IAA, adjudicate disagreements, freeze the gold-labeled dataset.
- **Single owner: Researcher A**
- **Deliverables:**
 - 100% double-annotated dataset
 - IAA report (overall + per-category κ)
 - Adjudication log with reasoning for every disagreement
 - Frozen gold dataset (v1.0), stratified splits (dev/test)
- **Required AI agents:** C4 (Legal Citation Validator, advisory support to annotators), C5 (QA/IAA — statistical computation).
- **AI tools:** Label Studio (annotation execution), Python/scikit-learn/statsmodels (κ computation, orchestrated via Claude Code scripts), Zotero (no new role here, dormant this phase).
- **Human supervision checkpoints:**
 - A manages annotator sessions directly (this phase is inherently human-labor-heavy, not agent-heavy).
 - A personally reviews every adjudication decision the senior legal expert makes, logging reasoning, not just the final label.

- A confirms $\kappa \geq 0.6$ overall and ≥ 0.5 per category before freezing — if not met, A alone decides whether to revise guidelines and re-pilot (a Phase 1 rollback) or accept a documented limitation (per Operating Manual, low κ blocks progression, so this is a real possible delay, planned for in the timeline buffer).
- **Entry criteria:** Phase 2 handoff received; annotators recruited and trained (from A's Phase 2 shadow-work); Label Studio project live.
- **Exit criteria:** All instances double-annotated; IAA thresholds met; gold set frozen and version-tagged (v1.0); splits generated and checksummed.
- **Estimated duration:** 4 weeks (longest phase — human annotation throughput is the bottleneck, not compute).
- **Git branch/workflow:** `phase3-annotation` branch, owned by A. Gold-label files are the most sensitive artifact in the repo — protect this branch with required review even though B isn't actively working here, since a silent gold-label error is the costliest possible mistake in the project.
- **Handoff artifacts to Phase 4:**
 - `gold_dataset_v1.0/` (DVC-tracked), `iaa_report.md`, `adjudication_log.md`, `splits/` with checksums
 - A short (20–30 min) handoff confirming any known caveats (e.g., a category with borderline κ that B should treat cautiously when interpreting per-category results later)

What B does during Phase 3 (independent, zero dependency on real gold labels): build and unit-test the full baseline evaluation harness (standard + agentic settings, per D1/D2) against synthetic placeholder data — every line of infrastructure code, prompt template, logging pipeline, and W&B integration can be written and tested now, so that the moment the real gold dataset lands, B can run real experiments immediately rather than spending week 1 of Phase 4 still building infrastructure. This is the single highest-leverage overlap in the whole plan, since Phase 3's 4 weeks would otherwise leave B idle for the longest stretch in the project.

PHASE 4 — EXPERIMENTATION & STATISTICAL VALIDATION

- **Objective:** Run all baseline models (standard + agentic), compute pre-registered statistical tests, produce the results matrix.
- **Single owner: Researcher B**
- **Deliverables:**
 - Full results matrix (≥ 4 models $\times$ 2 settings $\times$ task $\times$ category)
 - Statistical test outputs (effect sizes, CIs) for H1-H3
 - Error-analysis raw sample (stratified, ~ 50 instances) with preliminary technical failure-pattern notes
- **Required AI agents:** D1 (Experiment Design), D2 (Baseline Research), D3 (Statistical Analysis), D4 (Result Interpretation — raw sampling only at this stage, full interpretation deferred to the Phase 4 $\rightarrow$ 5 handoff).
- **AI tools:** Claude Code (harness execution, already built during Phase 3 shadow-work), Weights & Biases (run tracking), FAISS/Elasticsearch (agentic-setting retrieval index over the Phase 2 corpus).
- **Human supervision checkpoints:**
 - B personally approves exact prompt wording before any run (D1's protocol, locked before execution).
 - B flags any implausible run (near-0%/100%) for manual inspection rather than silently including it.
 - B confirms every statistical test matches the Stage 5 pre-registration exactly — no post-hoc test substitution.
- **Entry criteria:** Gold dataset v1.0 received from Phase 3; evaluation harness already built (from Phase 3 shadow-work), only needs pointing at real data.
- **Exit criteria:** Full results matrix complete, no unflagged failed runs; H1-H3 each have a reported effect size + CI; stratified error sample pulled and saved.
- **Estimated duration:** 2.5 weeks (shorter than it would otherwise be, precisely because the harness was pre-built during Phase 3's overlap window).

- **Git branch/workflow:** `phase4-experiments` branch, owned by B. Every run tagged with a git commit hash + W&B run ID, per the reproducibility requirement.
- **Handoff artifacts to Phase 5:**
 - `results_matrix.csv`, `stats_report.md`, `error_sample.json` (the full ~50-instance stratified sample, not just cherry-picked examples)
 - A 30-45 minute handoff walkthrough where B explains any surprising or ambiguous results directly to A, since A will need to interpret legal/linguistic meaning behind technical failure patterns when writing the Results/Error Analysis sections.

What A does during Phase 4 (independent, zero dependency on B's results): draft the Introduction, Related Work, Dataset/Taxonomy, and Annotation Methodology sections of the paper in full — none of these require experimental results, only Phase 1-3 artifacts A already owns. This is the second major overlap window, ensuring A isn't idle for 2.5 weeks and that Phase 5 starts with roughly half the paper already drafted.

PHASE 5 — SCIENTIFIC WRITING & DRAFT ASSEMBLY

- **Objective:** Produce the complete paper draft, integrating A's already-drafted sections (from Phase 4 overlap) with the Results/Experiments sections built from B's Phase 4 deliverables.
- **Single owner: Researcher A**
- **Deliverables:** Complete first full draft, every claim traced to an artifact (per E1's constraint), figures/tables finalized.
- **Required AI agents:** E1 (Scientific Writing), D4 (Result Interpretation — full pass now, turning B's raw error sample into the polished Error Analysis narrative).
- **AI tools:** Overleaf (shared LaTeX, though only A commits during this phase), NotebookLM (re-check any claim against the source papers while writing Related Work, to catch citation drift), Zotero (BibTeX export for the paper).

- **Human supervision checkpoints:**
 - A personally verifies every E1-flagged `[CLAIM NEEDS SOURCE]` marker is resolved before considering a section done.
 - A does not cherry-pick error-analysis examples — works from B's full stratified sample (handoff artifact from Phase 4).
- **Entry criteria:** Phase 4 handoff received; A's own Phase 4 overlap drafts already exist as a running start.
- **Exit criteria:** Full draft complete, <10% of claims lack a traceable artifact (per Operating Manual's Stage 21 threshold), all figures/tables finalized.
- **Estimated duration:** 2 weeks (shorter than it would be without the Phase 4 overlap head start).
- **Git branch/workflow:** `phase5-writing` branch (or Overleaf's native versioning if the team prefers it for the paper itself, with periodic syncs back to the git repo for citation/artifact linkage). A owns commits; B does not edit prose during this phase.
- **Handoff artifacts to Phase 6:**
 - Full draft PDF/LaTeX source
 - Claim-to-artifact mapping document
 - A short handoff flagging any sections where A is unsure a technical claim is stated correctly, for B to double-check during the reproducibility pass.

What B does during Phase 5 (independent, zero dependency on the draft's prose): reproducibility dry-run prep — pin all library versions, freeze exact model version/dates used, verify every result in `RESULTS.md` links to a real config/run, and begin the copyright/licensing review of every data source ahead of open-source release. None of this requires the paper text to exist yet.

PHASE 6 — REPRODUCIBILITY, RELEASE & REVIEWER SIMULATION

- **Objective:** Verify the full project is reproducible end-to-end, release the

dataset/code publicly, and run the final Reviewer Simulation pass against the complete draft.

- **Single owner: Researcher B**
- **Deliverables:**
 - Reproducibility dry-run completed successfully (someone other than the original pipeline author regenerates the main results table — ideally A attempts this using only B's documentation, which doubles as a genuine reproducibility test)
 - Public dataset + code release (HuggingFace + GitHub), DOI obtained (Zenodo)
 - Final Reviewer Simulation report (E2) against the complete Phase 5 draft
- **Required AI agents:** E2 (Reviewer Simulation) — full battery of 5 standard questions against the complete draft; A1 (Planner) coordinates final artifact linkage.
- **AI tools:** Claude Code (final repo cleanup, release packaging), Zenodo (DOI), HuggingFace Datasets Hub.
- **Human supervision checkpoints:**
 - B confirms copyright/licensing status is resolved for every released source (no full DLR/BLC/ALR text released, per earlier copyright guidance — metadata + fair-use excerpts only).
 - **Both A and B jointly resolve every E2 fatal-flaw flag** — this is the one place besides Phase 7 where joint work is required, since resolving a fatal flaw may require rewriting content (A's domain) informed by a technical constraint (B's domain).
- **Entry criteria:** Phase 5 draft received; RESULTS.md and reproducibility prep already largely complete (from Phase 5 overlap).
- **Exit criteria:** Reproducibility dry-run succeeds; public release live with DOI; all E2 fatal-flaw flags resolved and logged.
- **Estimated duration:** 1.5 weeks.

- **Git branch/workflow:** `phase6-release` branch, owned by B, merged to `main` and tagged as the submission-candidate commit.
 - **Handoff artifacts to Phase 7:** Final draft (post-flag-resolution), public release links, complete submission checklist (Operating Manual Part 8) with every box checked.
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PHASE 7 — FINAL JOINT REVIEW & SUBMISSION

- **Objective:** Final go/no-go decision and submission.
 - **Owner:** Joint — **unavoidably, per the Operating Manual's Part 0 rule that submission is a PI decision that cannot be delegated to a single track owner.**
 - **Deliverables:** Submitted paper.
 - **Required AI agents:** None new — this is a human decision gate.
 - **AI tools:** None required, beyond final formatting checks.
 - **Human supervision checkpoints:** Both researchers independently walk the full submission checklist; both must agree before submitting.
 - **Entry criteria:** Phase 6 exit criteria fully met.
 - **Exit criteria:** Submitted to ICCIT.
 - **Estimated duration:** 2–3 days.
 - **Git branch/workflow:** Merge `phase6-release` to `main`, tag `iccit-submission-v1`.
 - **Handoff artifacts:** None — this is the terminal phase.
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WHICH PHASES CAN SAFELY OVERLAP, AND WHICH MUST STAY STRICTLY SEQUENTIAL

Safe overlaps (deliberately designed into the plan above):

- Phase 1 (A) ↔ B's infra shadow-work — zero content dependency.
- Phase 2 (B) ↔ A's annotator recruitment/pilot — zero dependency on the actual corpus existing yet.
- Phase 3 (A) ↔ B's evaluation-harness construction against synthetic data — zero dependency on real gold labels.
- Phase 4 (B) ↔ A's early paper-section drafting — zero dependency on results.
- Phase 5 (A) ↔ B's reproducibility/licensing prep — zero dependency on final prose.

Must remain strictly sequential (no overlap possible):

- Phase 1 → Phase 2: B cannot scrape without a frozen scope and cannot build the fabrication pipeline without a frozen taxonomy.
- Phase 2 → Phase 3: A cannot annotate a corpus that doesn't exist yet.
- Phase 3 → Phase 4: B cannot run real experiments without real gold labels (synthetic-data testing in the overlap window is a dry run only, not real results).
- Phase 4 → Phase 5: A cannot write the Results section without real results.
- Phase 6 → Phase 7: submission cannot happen before reproducibility and fatal-flaw resolution are confirmed complete.

The rule of thumb: **overlap is safe exactly when the "shadow" work item does not consume or transform the active phase's actual output** — it only becomes unsafe the moment the shadow-worker would need to touch or depend on content the active owner hasn't finalized yet.

HOW TO PERFORM CLEAN HANDOFFS

A fixed, repeated protocol at every gate (Phases 1→2 through 6→7):

1. **Owner declares exit criteria met**, posts the handoff artifact list (per each phase's spec above) in the research log.

2. **Incoming owner reviews artifacts asynchronously first** (never walks in cold to the live session) — minimum 24 hours before the handoff meeting, to arrive with informed questions rather than discovering issues live.
3. **One handoff meeting, time-boxed** (30–60 minutes depending on phase complexity, specified per phase above) — outgoing owner walks incoming owner through decisions, edge cases, and known caveats. This is the *only* mandatory synchronous meeting between phases.
4. **Incoming owner confirms receipt in the decision log** with a dated entry: "Phase N handoff received, artifacts reviewed, no blocking questions" (or specifies what's blocking, which pauses the gate until resolved).
5. **Outgoing owner remains on-call but not co-working** for the first 2–3 days of the next phase — available for quick async questions, but not pulled back into active task ownership. This prevents the classic failure mode where handoffs are nominal but the outgoing owner ends up doing half the next phase anyway.

What makes a handoff "clean" rather than a disguised continuation of shared work: the incoming owner should be able to operate for the rest of the phase without needing the outgoing owner's ongoing involvement. If Phase 3 requires A to repeatedly ask B clarifying questions about the scraper throughout the month, the Phase 2 handoff was incomplete — the fix is a better handoff document/walkthrough, not an ongoing consultation arrangement.

HOW TO USE AGENTIC AI THROUGHOUT, WHILE THE HUMAN STAYS FINAL DECISION-MAKER

Unchanged from the Operating Manual's core rule, restated in single-owner terms: **within their phase, the sole owner is the human decision-maker for everything agents produce — there is no second researcher to defer to mid-phase, which makes the owner's individual discipline in verifying agent output more important, not less.** Concretely:

- Every agent invocation inside a phase is logged by the owner alone (no shared review burden, but also no second pair of eyes until the handoff review) — this raises the bar on the owner personally following the evidence-verification rules in Operating Manual Part 0, since nobody else will catch a lapse mid-phase.
 - The handoff review (step 2 above) doubles as the delayed "second pair of eyes" — the incoming owner should specifically check that agent-derived claims/artifacts from the outgoing phase have real evidence trails, since this is the first point anyone besides the phase owner examines that work closely.
 - The two mandatory *joint* human checkpoints that survive this single-owner redesign (fabrication-batch content review in Phase 2, and fatal-flaw resolution in Phase 6) are joint specifically because they are the two points where an agent's output has the highest chance of encoding a subtle content error (a too-easy/too-hard fabrication, or an unresolved reviewer-flagged gap) that a single owner working alone is more likely to miss.
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HOW TO MAXIMIZE PRODUCTIVITY AND QUALITY WITH ONLY TWO RESEARCHERS

1. **Never let a researcher be simultaneously idle and blocked** — every phase has a designed shadow-work assignment for the non-owner, so total person-weeks of the ~15.5-week timeline are closer to 2× utilized than a naive sequential read of "15.5 weeks × 1 person" would suggest.
2. **Keep the number of synchronous meetings to exactly the handoff gates** (6 of them, ~30–60 minutes each) plus Phase 7's short joint session — everything else is asynchronous artifact review, which respects deep-work time for both researchers during their owned phases.
3. **Front-load the highest-uncertainty phase (Phase 1) with the person best equipped to resolve it fast** — A's novelty/gap judgment is the single biggest risk to

the whole timeline, so it goes first, single-owned, with no engineering distractions competing for A's attention that week.

4. **Protect the gold-label branch (Phase 3) with review requirements even though B isn't actively co-working it** — the one place where "no oversight" is not an acceptable tradeoff for coordination-cost savings, since a silent gold-label defect propagates through every later phase undetected until it's expensive to fix.
5. **Build infrastructure before you need it, not when you need it** — the Phase 1 and Phase 3 shadow-work windows exist specifically so that when B needs the evaluation harness in Phase 4, it's already tested against synthetic data and Phase 4 becomes "point at real data and run" rather than "build and debug under deadline pressure."